

### §3. Conjugate Direction Methods

Conjugate direction methods can be regarded as being somewhat intermediate between the steepest descent method and Newton's method. They are motivated by the desire to accelerate the typically slow convergence associated with the steepest descent method while avoiding the information requirement associated with the evaluation, storage, and inversion of the Hessian as required by Newton's Method.

Conjugate direction methods are considered among the best general purpose methods presently available.

**Definition 3.1.** *Given a symmetric matrix  $Q$ , two vectors  $\mathbf{d}_1$  and  $\mathbf{d}_2$  are said to be  $Q$ -orthogonal, or conjugate with respect to  $Q$ , if  $\mathbf{d}_1^T Q \mathbf{d}_2 = 0$ .*

*A finite set of vectors  $\mathbf{d}_0, \mathbf{d}_1, \dots, \mathbf{d}_k$  is said to be  $Q$ -orthogonal if  $\mathbf{d}_i^T Q \mathbf{d}_j = 0$  for all  $i \neq j$ .*

**Theorem 3.1.** *If  $Q$  is positive definite and the set of nonzero vectors  $\mathbf{d}_0, \mathbf{d}_1, \dots, \mathbf{d}_k$  is  $Q$ -orthogonal, then these vectors are linearly independent.*

*Proof.* Let  $\alpha_0, \alpha_1, \dots, \alpha_k$  be any constants such that

$$\alpha_0 \mathbf{d}_0 + \alpha_1 \mathbf{d}_1 + \dots + \alpha_k \mathbf{d}_k = \mathbf{0}.$$

Multiplying by  $Q$  and taking the inner product with  $\mathbf{d}_i$  yields  $\alpha_i \mathbf{d}_i^T Q \mathbf{d}_i = 0$ . Since  $Q$  is positive definite,  $\mathbf{d}_i^T Q \mathbf{d}_i > 0$ . It follows that  $\alpha_i = 0$ .  $\square$

Conjugate direction methods were originally invented and analyzed for the pure quadratic problem

$$\text{Min } \frac{1}{2} \mathbf{x}^T Q \mathbf{x} - \mathbf{b}^T \mathbf{x} \quad (3.1)$$

where  $Q$  is an  $n \times n$  positive definite matrix. Note that the unique solution to (3.1) is also the unique solution to

$$Q \mathbf{x} = \mathbf{b}. \quad (3.2)$$

REMARK. The detailed analysis of the pure quadratic problem can lead to significant practical advances. Indeed, conjugate direction methods, especially the method of conjugate gradients, have proved to be extremely effective in dealing with general objective functions.

**Theorem 3.2.** *(Conjugate Direction Theorem – Quadratic Case)*

*Let  $\{\mathbf{d}_i\}_{i=0}^{n-1}$  be a set of nonzero  $Q$ -orthogonal vectors. For any  $\mathbf{x}_0 \in \mathbb{R}^n$ , the sequence generated according to*

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k, \quad k \geq 0 \quad (3.3)$$

with

$$\alpha_k = -\frac{\mathbf{g}_k^T \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k} \quad (3.4)$$

and

$$\mathbf{g}_k = Q\mathbf{x}_k - \mathbf{b} \quad (3.5)$$

converges to the unique solution  $\mathbf{x}^*$  of  $Q\mathbf{x} = \mathbf{b}$  after  $n$  steps, that is,  $\mathbf{x}_n = \mathbf{x}^*$ .

*Proof.* Since  $\mathbf{d}'_k$ s are linearly independent, we can write

$$\mathbf{x}^* - \mathbf{x}_0 = \alpha_0 \mathbf{d}_0 + \alpha_1 \mathbf{d}_1 + \cdots + \alpha_{n-1} \mathbf{d}_{n-1} \quad (3.6)$$

for some constants  $\alpha'_k$ s. Now set  $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k$  for  $k \geq 0$ . Then

$$\begin{aligned} \mathbf{x}_k &= \mathbf{x}_{k-1} + \alpha_{k-1} \mathbf{d}_{k-1} \\ &= \mathbf{x}_{k-2} + \alpha_{k-2} \mathbf{d}_{k-2} + \alpha_{k-1} \mathbf{d}_{k-1} \\ &= \cdots \\ &= \mathbf{x}_0 + \alpha_0 \mathbf{d}_0 + \alpha_1 \mathbf{d}_1 + \cdots + \alpha_{k-1} \mathbf{d}_{k-1}, \end{aligned}$$

which implies that

$$\mathbf{x}_k - \mathbf{x}_0 = \alpha_0 \mathbf{d}_0 + \alpha_1 \mathbf{d}_1 + \cdots + \alpha_{k-1} \mathbf{d}_{k-1}. \quad (3.7)$$

By (3.6) and (3.7) we get  $\mathbf{x}^* - \mathbf{x}_0 = \mathbf{x}_n - \mathbf{x}_0$ , so  $\mathbf{x}_n = \mathbf{x}^*$ . Hence to prove the theorem, it suffices to prove that  $\alpha_k$  is as given in (3.4). For this purpose, multiplying (3.6) by  $Q$  and taking the inner product with  $\mathbf{d}_k$  gives

$$\mathbf{d}_k^T Q(\mathbf{x}^* - \mathbf{x}_0) = \alpha_k \mathbf{d}_k^T Q \mathbf{d}_k.$$

So

$$\alpha_k = \frac{\mathbf{d}_k^T Q(\mathbf{x}^* - \mathbf{x}_0)}{\mathbf{d}_k^T Q \mathbf{d}_k}. \quad (3.8)$$

Using (3.7), we obtain

$$\begin{aligned} \mathbf{d}_k^T Q(\mathbf{x}_k - \mathbf{x}_0) &= \mathbf{d}_k^T Q[\alpha_0 \mathbf{d}_0 + \alpha_1 \mathbf{d}_1 + \cdots + \alpha_{k-1} \mathbf{d}_{k-1}] \\ &= 0. \end{aligned}$$

Substituting it into (3.8) produces

$$\alpha_k = \frac{\mathbf{d}_k^T Q(\mathbf{x}^* - \mathbf{x}_k)}{\mathbf{d}_k^T Q \mathbf{d}_k} = -\frac{\mathbf{g}_k^T \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k},$$

completing the proof. □

EXAMPLE 3.1. (i) Let  $Q$  be a symmetric matrix. Show that any two eigenvectors of  $Q$ , corresponding to distinct eigenvalues, are  $Q$ -conjugate.

(ii) Using the conjugate direction theorem, solve the problem

$$\text{Min } f(x_1, x_2) = x_1^2 + x_2^2 - x_1x_2 + x_1 - 2x_2. \quad (3.9)$$

SOLUTION. The proof of (i) is left as an exercise. To solve (ii), let us first transform (3.9) into the standard form (3.1), which yields  $Q = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ ,  $\mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ . It is easy to see that  $\mathbf{d}_0 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$  and  $\mathbf{d}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$  are two eigenvectors of  $Q$ , corresponding to eigenvalues 3 and 1, respectively. According to statement (i),  $\mathbf{d}_0$  and  $\mathbf{d}_1$  are  $Q$ -conjugate. Thus we may apply the conjugate direction theorem. Choose, for instance,  $\mathbf{x}_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ . Then by direction computation, we have  $\mathbf{x}_1 = \begin{pmatrix} -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$  and  $\mathbf{x}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ . By virtue of Theorem 3.2,  $\mathbf{x}_2$  is the optimal solution.  $\square$

The conjugate direction theorem is based on the hypothesis that  $\{\mathbf{d}_i\}_{i=0}^{n-1}$  are specified beforehand, a natural question is: how can we get these conjugate directions?

### The Conjugate Gradient Method

The conjugate gradient method is the conjugate direction method that is obtained by selecting the successive direction vectors as a conjugate version of the successive gradients obtained as the method progresses. Thus, the directions are not specified beforehand, but rather are determined sequentially at each step of the iteration. At each step, one evaluates the current negative gradient vector and adds to it a linear combination of the previous direction vectors to obtain a new conjugate direction vector along which to move.

#### Method 1. (Conjugate Gradient Algorithm – Quadratic Case)

Starting at any  $\mathbf{x}_0$  define  $\mathbf{d}_0 = -\mathbf{g}_0 = \mathbf{b} - Q\mathbf{x}_0$  and

$$\begin{aligned} \mathbf{x}_{k+1} &= \mathbf{x}_k + \alpha_k \mathbf{d}_k \\ \alpha_k &= -\frac{\mathbf{g}_k^T \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k} \\ \mathbf{d}_{k+1} &= -\mathbf{g}_{k+1} + \beta_k \mathbf{d}_k \\ \beta_k &= \frac{\mathbf{g}_{k+1}^T Q \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k} \quad \text{where } \mathbf{g}_k = Q\mathbf{x}_k - \mathbf{b} \end{aligned}$$

**Theorem 3.2.** (*Conjugate Gradient Theorem – Quadratic Case*)

The conjugate gradient algorithm is a conjugate direction method, that is, the vectors  $\{\mathbf{d}_k\}$ 's are  $Q$ -orthogonal.

*Proof.* Omitted. □

REMARK 1. It can be guaranteed that at some step  $k(\leq n)$ ,  $\mathbf{g}_k = 0$ . Whenever this step is reached, the process terminates and  $\mathbf{x}_k$  is the optimal solution.

REMARK 2. There are many methods for generating conjugate directions, among them the conjugate gradient method is most important, because

- (a) Unless the solution is obtained in less than  $n$  steps, the gradient is always nonzero and linearly independent of all previous direction vectors.
- (b) The formula is simple.
- (c) The directions are based on the gradient, the process makes good uniform progress toward the solution at every step. It is important for generalizations to nonquadratic problems.

EXAMPLE 3.2. Solve the following problem using the conjugate gradient method

$$\text{Min } f(x_1, x_2) = \frac{3}{2}x_1^2 + \frac{1}{2}x_2^2 - x_1x_2 - 2x_1. \quad (3.10)$$

SOLUTION. Write (3.10) in the standard form, we get  $Q = \begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix}$ ,  $\mathbf{b} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ . Now let us start with  $\mathbf{x}_0 = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$ . Since  $\nabla f(\mathbf{x}) = Q\mathbf{x} - \mathbf{b} = \begin{pmatrix} 3x_1 - x_2 - 2 \\ -x_1 + x_2 \end{pmatrix}$ , we have  $\mathbf{g}_0 = \nabla f(\mathbf{x}_0) = \begin{pmatrix} -12 \\ 6 \end{pmatrix}$  and  $\mathbf{d}_0 = \begin{pmatrix} 12 \\ -6 \end{pmatrix}$ . It follows that  $\alpha_0 = \frac{5}{17}$ , and thus  $\mathbf{x}_1 = \mathbf{x}_0 + \alpha_0\mathbf{d}_0 = \begin{pmatrix} \frac{26}{17} \\ \frac{38}{17} \end{pmatrix}$ . Next, it is easy to compute that  $\mathbf{g}_1 = \nabla f(\mathbf{x}_1) = \begin{pmatrix} \frac{6}{17} \\ \frac{12}{17} \end{pmatrix}$ , and  $\beta_0 = \frac{1}{289}$ . So  $\mathbf{d}_1 = -\mathbf{g}_1 + \beta_0\mathbf{d}_0 = \begin{pmatrix} -\frac{90}{289} \\ -\frac{210}{289} \end{pmatrix}$ , and  $\alpha_1 = \frac{17}{10}$ . Hence,  $\mathbf{x}_2 = \mathbf{x}_1 + \alpha_1\mathbf{d}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ . Notice that  $\mathbf{g}_2 = \nabla f(\mathbf{x}_2) = 0$ , so  $\mathbf{x}_2$  is the optimal solution. □

Quadratic Approximations for Nonquadratic Problems

The general unconstrained problem on  $\mathbb{R}^n$

$$\text{Minimize } f(\mathbf{x})$$

can be attacked by making suitable approximations to the conjugate gradient method. We make the following associations at  $\mathbf{x}_k$ :

$$\mathbf{g}_k = \nabla f(\mathbf{x}_k), \quad Q = \nabla^2 f(\mathbf{x}_k)$$

and using these associations, reevaluated at each step, all quantities necessary to implement the basic conjugate gradient algorithm can be evaluated.

Method 2. (Conjugate Gradient Algorithm – Nonquadratic Case)

Step 1. Starting at  $\mathbf{x}_0$  compute  $\mathbf{g}_0 = \nabla f(\mathbf{x}_0)$  and set  $\mathbf{d}_0 = -\mathbf{g}_0$ .

Step 2. For  $k = 0, 1, \dots, n-1$ , do:

- (a) Set  $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k$ , where  $\alpha_k = \frac{-\mathbf{g}_k^T \mathbf{d}_k}{\mathbf{d}_k^T \nabla^2 f(\mathbf{x}_k) \mathbf{d}_k}$ .
- (b) Compute  $\mathbf{g}_{k+1} = \nabla f(\mathbf{x}_{k+1})$ .
- (c) Unless  $k = n-1$ , set  $\mathbf{d}_{k+1} = -\mathbf{g}_{k+1} + \beta_k \mathbf{d}_k$ , where  $\beta_k = \frac{\mathbf{g}_{k+1}^T \nabla^2 f(\mathbf{x}_k) \mathbf{d}_k}{\mathbf{d}_k^T \nabla^2 f(\mathbf{x}_k) \mathbf{d}_k}$  and repeat (a).

Step 3. Replace  $\mathbf{x}_0$  by  $\mathbf{x}_n$  and go back to Step 1.

REMARK. The termination criterion may be chosen as  $\|\mathbf{g}_k\| < \varepsilon$  for some  $\varepsilon > 0$  specified beforehand.

#### §4. Constrained Minimization Conditions

Consider general optimization problems of the form

$$\begin{array}{ll} \text{Minimize} & f(\mathbf{x}) \\ \text{subject to} & h_1(\mathbf{x}) = 0 \quad g_1(\mathbf{x}) \leq 0 \\ & h_2(\mathbf{x}) = 0 \quad g_2(\mathbf{x}) \leq 0 \\ & \vdots \quad \vdots \\ & h_m(\mathbf{x}) = 0 \quad g_q(\mathbf{x}) \leq 0 \\ & \mathbf{x} \in \Omega \subseteq \mathbb{R}^n \end{array}$$

where the functions  $f$ , each  $h_i$  and each  $g_j$  are continuous, and usually assumed to be in  $\mathcal{C}^2$ . For simplicity, we introduce the vector-valued functions  $\mathbf{h} = (h_1, h_2, \dots, h_m)$  and  $\mathbf{g} = (g_1, g_2, \dots, g_q)$ , rewrite the original form as

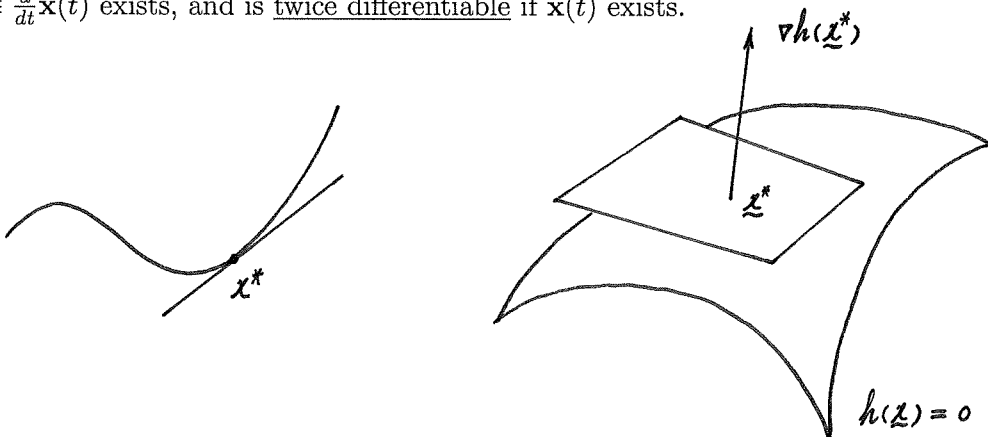
$$\begin{array}{ll} \text{Minimize} & f(\mathbf{x}) \\ \text{subject to} & \mathbf{h}(\mathbf{x}) = 0 \\ & \mathbf{g}(\mathbf{x}) \leq 0 \\ & \mathbf{x} \in \Omega \subseteq \mathbb{R}^n \end{array} \quad (4.1)$$

The constraints  $\mathbf{h}(\mathbf{x}) = 0$ ,  $\mathbf{g}(\mathbf{x}) \leq 0$  are referred to as functional constraints, while the constraint  $\mathbf{x} \in \Omega$  is a set constraint. An inequality constraint  $g_i(\mathbf{x}) \leq 0$  is said to be active at a feasible point  $\mathbf{x}$  if  $g_i(\mathbf{x}) = 0$  and inactive at  $\mathbf{x}$  if  $g_i(\mathbf{x}) < 0$ . We refer to any equality constraint  $h_i(\mathbf{x}) = 0$  as active at any feasible point.

A surface  $S$  in  $\mathbb{R}^n$  is defined by a set of equality constraints

$$\begin{cases} h_1(\mathbf{x}) = 0 \\ h_2(\mathbf{x}) = 0 \\ \vdots \\ h_m(\mathbf{x}) = 0. \end{cases} \quad (4.2)$$

We say that  $S$  is smooth if each  $h_i \in \mathcal{C}^1$ . A curve on a surface  $S$  is a family of points  $\mathbf{x}(t) \in S$  continuously parameterized by  $t$  for  $a \leq t \leq b$ . The curve is differentiable if  $\dot{\mathbf{x}} \equiv \frac{d}{dt}\mathbf{x}(t)$  exists, and is twice differentiable if  $\ddot{\mathbf{x}}(t)$  exists.



A curve is said to pass through the point  $\mathbf{x}^*$  if  $\mathbf{x}^* = \mathbf{x}(t^*)$  for some  $t^*$  with  $a < t^* < b$ . The derivative of the curve at  $\mathbf{x}^*$  is defined as  $\dot{\mathbf{x}}(t^*)$ . Consider all differentiable curves on

$S$  passing through a point  $\mathbf{x}^*$ . The tangent plane at  $\mathbf{x}^*$  is defined as the collection of the derivatives of all these differentiable curves at  $\mathbf{x}^*$ .

The problem of obtaining an explicit representation for the tangent plane is a fundamental problem.

**Definition 4.1.** A point  $\mathbf{x}^*$  satisfying (4.2) is said to be a regular point of constraint if  $\nabla h_1(\mathbf{x}^*), \nabla h_2(\mathbf{x}^*), \dots, \nabla h_m(\mathbf{x}^*)$  are linearly independent.

**Theorem 4.1.** At a regular point  $\mathbf{x}^*$  of the surface  $S$  defined by  $\mathbf{h}(\mathbf{x}) = 0$ , the tangent plane is equal to

$$M = \{\mathbf{y} \in \mathbb{R}^n : \nabla \mathbf{h}(\mathbf{x}^*)^T \mathbf{y} = 0\}, \quad \text{where} \quad \nabla \mathbf{h}(\mathbf{x}) = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_m}{\partial x_1} \\ \vdots & & \vdots \\ \frac{\partial h_1}{\partial x_n} & \cdots & \frac{\partial h_m}{\partial x_n} \end{bmatrix}.$$

*Proof.* Omitted.

**Lemma 4.1.** Let  $\mathbf{x}^*$  be a regular point of constraints  $\mathbf{h}(\mathbf{x}) = 0$  and a local extremum point of  $f$  subject to these constraints. Then all  $\mathbf{y} \in \mathbb{R}^n$  satisfying

$$\nabla \mathbf{h}(\mathbf{x}^*)^T \mathbf{y} = 0$$

must also satisfy  $\nabla f(\mathbf{x}^*)^T \mathbf{y} = 0$ .

*Proof.* Let  $\mathbf{y}$  be a point in  $\mathbb{R}^n$  with  $\nabla \mathbf{h}(\mathbf{x}^*)^T \mathbf{y} = 0$ . Then, by Theorem 4.1,  $\mathbf{y}$  is a vector in the tangent plane at  $\mathbf{x}^*$ . Thus there exists a smooth curve  $\mathbf{x}(t)$  on the constraint surface such that  $\mathbf{x}(0) = \mathbf{x}^*$ ,  $\dot{\mathbf{x}}(0) = \mathbf{y}$ , and  $\mathbf{h}(\mathbf{x}(t)) = 0$  for  $-a \leq t \leq a$ , for some  $a > 0$ .

Since  $\mathbf{x}^*$  is a local extremum point, 0 is a local extremum point of  $f(\mathbf{x}(t))$ . So we have

$$\left. \frac{d}{dt} f(\mathbf{x}(t)) \right|_{t=0} = 0$$

or  $\nabla f(\mathbf{x}^*)^T \mathbf{y} = 0$ . □

**Theorem 4.2.** (First - Order Necessary Conditions)

Let  $\mathbf{x}^*$  be a local extremum point of  $f$  subject to the constraints  $\mathbf{h}(\mathbf{x}) = 0$ . Assume further that  $\mathbf{x}^*$  is a regular point of these constraints. Then there is a  $\boldsymbol{\lambda} \in \mathbb{R}^m$  such that

$$\nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*) \boldsymbol{\lambda} = 0.$$

*Proof.* From Lemma 4.1, it can be seen that  $\nabla f(\mathbf{x}^*)^T \mathbf{y} = 0$  whenever  $\nabla \mathbf{h}(\mathbf{x}^*)^T \mathbf{y} = 0$ . So the two systems  $\nabla \mathbf{h}(\mathbf{x}^*)^T \mathbf{y} = 0$  and  $[\nabla \mathbf{h}(\mathbf{x}^*) \mid \nabla f(\mathbf{x}^*)]^T \mathbf{y} = 0$  have the same solution

set. Hence  $\text{rk} [\nabla \mathbf{h}(\mathbf{x}^*) \mid \nabla f(\mathbf{x}^*)] = \text{rk} [\nabla \mathbf{h}(\mathbf{x}^*)]$ , which implies that  $\nabla f(\mathbf{x}^*) = \nabla \mathbf{h}(\mathbf{x}^*)\boldsymbol{\mu}$  or  $\nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*)\boldsymbol{\lambda} = 0$ .  $\square$

REMARK.  $\boldsymbol{\lambda}$  is usually called the Lagrange multiplier. Note that  $\nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*)\boldsymbol{\lambda} = 0$  together with  $\mathbf{h}(\mathbf{x}^*) = 0$  give a total of  $n + m$  equations in  $n + m$  variables comprising  $\mathbf{x}^*$  and  $\boldsymbol{\lambda}$ . They are sufficient to get a unique local extremum.

EXAMPLE 4.1. Consider the problem

$$\begin{array}{ll} \text{Min} & x_1x_2 + x_2x_3 + x_3x_1 \\ \text{s.t.} & x_1 + x_2 + x_3 = 3. \end{array}$$

SOLUTION. Since  $f(\mathbf{x}) = x_1x_2 + x_2x_3 + x_3x_1$  and  $h(\mathbf{x}) = x_1 + x_2 + x_3 - 3$ , we have

$$\nabla f(\mathbf{x}) = \begin{bmatrix} x_2 + x_3 \\ x_3 + x_1 \\ x_1 + x_2 \end{bmatrix}, \quad \nabla h(\mathbf{x}) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

According to the necessary condition,  $\nabla f(\mathbf{x}^*) + \nabla h(\mathbf{x}^*)\boldsymbol{\lambda} = 0$ . So

$$\begin{bmatrix} x_2 + x_3 \\ x_3 + x_1 \\ x_1 + x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \boldsymbol{\lambda} = 0.$$

Combining all these equations with  $x_1 + x_2 + x_3 = 3$ , we obtain  $(x_1, x_2, x_3, \boldsymbol{\lambda}) = (1, 1, 1, -2)$ .

**Theorem 4.3.** (*Second-Order Necessary Conditions*)

Let  $f, \mathbf{h} \in \mathcal{C}^2$ . Suppose that  $\mathbf{x}^*$  is a local minimum of  $f$  subject to  $\mathbf{h}(\mathbf{x}) = 0$  and that  $\mathbf{x}^*$  is a regular point of these constraints. Then there exists  $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \dots, \lambda_m)^T \in \mathbb{R}^m$  such that

$$\nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*)\boldsymbol{\lambda} = 0. \quad (4.3)$$

Moreover,  $L(\mathbf{x}^*) = \nabla^2 f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \nabla^2 h_i(\mathbf{x}^*)$  is positive semi-definite for all  $\mathbf{y}$  in  $\{\mathbf{y} : \nabla \mathbf{h}(\mathbf{x}^*)^T \mathbf{y} = 0\}$ .

*Proof.* Let  $\mathbf{x}(t)$  be twice differentiable curve on the surface defined by  $\mathbf{h}(\mathbf{x}) = 0$  with  $\mathbf{x}(0) = \mathbf{x}^*$  and  $\dot{\mathbf{x}}(0) = \mathbf{y}$ , and let  $g(t) \triangleq f(\mathbf{x}(t))$ . Then  $g(h) - 2g(0) + g(-h) \geq 0$  for sufficiently small  $h$  as  $\mathbf{x}^*$  is a local minimum point. Hence by L'Hospital rule, we have

$$g''(0) = \lim_{h \rightarrow 0} \frac{g(h) - 2g(0) + g(-h)}{h^2} \geq 0.$$

It follows that

$$\left. \frac{d^2}{dt^2} f(\mathbf{x}(t)) \right|_{t=0} = g''(0) \geq 0.$$



Thus

$$\dot{\mathbf{x}}(0)^T \nabla^2 f(\mathbf{x}^*) \dot{\mathbf{x}}(0) + \nabla f(\mathbf{x}^*)^T \ddot{\mathbf{x}}(0) \geq 0. \quad (4.4)$$

Differentiate  $\sum_{i=1}^m \lambda_i h_i(\mathbf{x}(t)) = 0$  with respect to  $t$  twice, we have

$$\sum_{i=1}^m \lambda_i \dot{\mathbf{x}}(0)^T \nabla^2 h_i(\mathbf{x}^*) \dot{\mathbf{x}}(0) + \sum_{i=1}^m \lambda_i \nabla h_i(\mathbf{x}^*)^T \ddot{\mathbf{x}}(0) = 0 \quad (4.5)$$

Add (4.4) to (4.5), in view of (4.3), we have

$$\dot{\mathbf{x}}(0)^T \left[ \sum_{i=1}^m \lambda_i \nabla^2 h_i(\mathbf{x}^*) + \nabla^2 f(\mathbf{x}^*) \right] \dot{\mathbf{x}}(0) \geq 0$$

or  $\mathbf{y}^T L(\mathbf{x}^*) \mathbf{y} \geq 0$ . □

**Theorem 4.4.** (*Second-Order Sufficient Conditions*)

Let  $f, \mathbf{h} \in \mathcal{C}^2$  and let  $\mathbf{x}^* \in \mathbb{R}^n, \boldsymbol{\lambda} \in \mathbb{R}^m$  be such that

- (i)  $\mathbf{h}(\mathbf{x}^*) = 0$ ;
- (ii)  $\nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*) \boldsymbol{\lambda} = 0$ ;
- (iii)  $L(\mathbf{x}^*) = \nabla^2 f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \nabla^2 h_i(\mathbf{x}^*)$  is positive definite on  $\{\mathbf{y} : \nabla \mathbf{h}(\mathbf{x}^*)^T \mathbf{y} = 0\}$ .

Then  $\mathbf{x}^*$  is a strict local minimum of  $f$  subject to  $\mathbf{h}(\mathbf{x}) = 0$ . □

EXAMPLE 4.2. Solve the problem

$$\begin{array}{ll} \text{Max} & x_1 x_2 + x_2 x_3 + x_3 x_1 \\ \text{s.t.} & x_1 + x_2 + x_3 = 3. \end{array}$$

SOLUTION. Based on the first-order necessary condition, we obtain  $(x_1^*, x_2^*, x_3^*, \lambda) = (1, 1, 1, -2)$  (recall Example 4.1). Note that

$$L = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} + [0] = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix},$$

which is neither positive nor negative definite. However, on the tangent plane  $M = \{\mathbf{y} : \nabla \mathbf{h}(\mathbf{x}^*)^T \mathbf{y} = 0\} = \{\mathbf{y} : y_1 + y_2 + y_3 = 0\}$ , we have

$$\begin{aligned} \mathbf{y}^T L(\mathbf{x}^*) \mathbf{y} &= (y_1, y_2, y_3) \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\ &= (-y_1, -y_2, -y_3) \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \\ &= -(y_1^2 + y_2^2 + y_3^2). \end{aligned}$$

So  $L$  is negative definite on  $M$ . Hence, by Theorem 4.4,  $\mathbf{x}^*$  is a strict local maximum. □

**Definition 4.2.** Let  $\mathbf{x}^*$  be a point satisfying the constraints

$$\mathbf{h}(\mathbf{x}) = 0 \quad \text{and} \quad \mathbf{g}(\mathbf{x}) \leq 0 \quad (4.6)$$

and let  $J$  be the set of indices  $j$  for which  $g_j(\mathbf{x}^*) = 0$ . Then  $\mathbf{x}^*$  is said to be a regular point of the constraint (4.6) if the gradient vectors  $\nabla h_i(\mathbf{x}^*), \nabla g_j(\mathbf{x}^*), 1 \leq i \leq m, j \in J$  are linearly independent.

**Theorem 4.5.** (Kuhn-Tucker Conditions)

Let  $f, \mathbf{g}, \mathbf{h} \in C^1$  and let  $\mathbf{x}^*$  be a local minimum point of the problem

$$\begin{aligned} \text{Min} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{h}(\mathbf{x}) = 0 \\ & \mathbf{g}(\mathbf{x}) \leq 0. \end{aligned}$$

and a regular point for the constraints. Then there exist  $\boldsymbol{\lambda} \in \mathbb{R}^m$  and  $\boldsymbol{\mu} \in \mathbb{R}^q$  with  $\boldsymbol{\mu} \geq 0$  such that

$$\begin{aligned} \nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*)\boldsymbol{\lambda} + \nabla \mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} &= 0 \\ \mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} &= 0 \end{aligned}$$

*Proof.* Let  $J$  be the set of indices  $j$  for which  $g_j(\mathbf{x}^*) = 0$ . Then  $\mathbf{x}^*$  is also a local minimum of the problem (why?)

$$\begin{aligned} \text{Min} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{h}(\mathbf{x}) = 0 \\ & g_j(\mathbf{x}) = 0 \quad \forall j \in J. \end{aligned}$$

By the first-order necessary condition, there exist  $\boldsymbol{\lambda} \in \mathbb{R}^m$  and  $\mu_j, j \in J$ , such that

$$\nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*)\boldsymbol{\lambda} + \sum_{j \in J} \mu_j \nabla g_j(\mathbf{x}^*) = 0. \quad (4.7)$$

Define  $\mu_j = 0$  if  $j \in \{1, 2, \dots, q\} - J$ . Then we have

$$\begin{aligned} \nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*)\boldsymbol{\lambda} + \nabla \mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} &= 0 \\ \mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} &= 0. \end{aligned}$$

The rest is to show that  $\mu_j \geq 0$  for each  $j \in J$ .

Suppose to the contrary that  $\mu_k < 0$  for some  $k \in J$ . Let  $S$  stand for the surface  $\mathbf{h}(\mathbf{x}) = 0, g_j(\mathbf{x}) = 0 \forall j \in J - \{k\}$ , and let  $M$  stand for  $\{\mathbf{y} : \nabla \mathbf{h}(\mathbf{x}^*)^T \mathbf{y} = 0, \nabla g_j(\mathbf{x}^*)^T \mathbf{y} = 0 \text{ for all } j \in J - \{k\}\}$ . Then there exists  $\mathbf{y} \in M$  such that  $\nabla g_k(\mathbf{x}^*)^T \mathbf{y} < 0$ . From (4.7), it follows that  $\nabla f(\mathbf{x}^*)^T \mathbf{y} < 0$ . Let  $\mathbf{x}(t)$  be a continuously differentiable curve on  $S$  passing through  $\mathbf{x}^*$  with  $\mathbf{x}(0) = \mathbf{x}^*$  and  $\dot{\mathbf{x}}(0) = \mathbf{y}$ . Then for small  $t \geq 0$ ,  $\mathbf{x}(t)$  is feasible, and

$$\left. \frac{d}{dt} f(\mathbf{x}(t)) \right|_{t=0} = \nabla f(\mathbf{x}^*)^T \mathbf{y} < 0,$$

contradicting the minimality of  $\mathbf{x}^*$ . □

REMARK. Notice that  $\mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} = 0$  iff  $\mu_i g_i(\mathbf{x}^*) = 0$  for all  $i$ .

**Theorem 4.6.** (*Second-Order Necessary Conditions*)

Let  $f, \mathbf{g}, \mathbf{h} \in \mathcal{C}^2$  and let  $\mathbf{x}^*$  be a regular point of constraints  $\mathbf{h}(\mathbf{x}) = 0$ ,  $\mathbf{g}(\mathbf{x}) \leq 0$  and a local minimum point of  $f$  subject to these constraints. Then there exist  $\boldsymbol{\lambda} \in \mathbb{R}^m$  and  $\boldsymbol{\mu} \in \mathbb{R}^q$  with  $\mu \geq 0$  such that

$$\begin{aligned}\nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*)\boldsymbol{\lambda} + \nabla \mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} &= 0 \\ \mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} &= 0.\end{aligned}$$

Moreover,  $L(\mathbf{x}^*) = \nabla^2 f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \nabla^2 h_i(\mathbf{x}^*) + \sum_{j=1}^q \mu_j \nabla^2 g_j(\mathbf{x}^*)$  is positive semi-definite on the tangent subspace of the active constraints at  $\mathbf{x}^*$ .

*Proof.* Since  $\mathbf{x}^*$  is a local minimum point over the constraints  $\mathbf{h}(\mathbf{x}) = 0$  and  $\mathbf{g}(\mathbf{x}) \leq 0$ , it is also a local minimum point for the problem with the active constraints taken as equality constraints. So the original problem can be replaced by the corresponding problem having equality constraints only.  $\square$

**Theorem 4.7.** (*Second-Order Sufficient Conditions*)

Let  $f, \mathbf{g}, \mathbf{h} \in \mathcal{C}^2$  and let  $\mathbf{x}^* \in \mathbb{R}^n, \boldsymbol{\lambda} \in \mathbb{R}^m, \boldsymbol{\mu} \in \mathbb{R}^q$  be such that

- (i)  $\mathbf{h}(\mathbf{x}^*) = 0, \quad \mathbf{g}(\mathbf{x}^*) \leq 0;$
- (ii)  $\boldsymbol{\mu} \geq 0;$
- (iii)  $\mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} = 0;$
- (iv)  $\nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*)\boldsymbol{\lambda} + \nabla \mathbf{g}(\mathbf{x}^*)\boldsymbol{\mu} = 0;$
- (v)  $L(\mathbf{x}^*) = \nabla^2 f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \nabla^2 h_i(\mathbf{x}^*) + \sum_{j=1}^q \mu_j \nabla^2 g_j(\mathbf{x}^*)$  is positive definite on the subspace

$$M = \{\mathbf{y} : \nabla \mathbf{h}(\mathbf{x}^*)^T \mathbf{y} = 0, \nabla g_j(\mathbf{x}^*)^T \mathbf{y} = 0 \quad \forall j \in J\}$$

where  $J = \{j : g_j(\mathbf{x}^*) = 0, \mu_j > 0\}$ .

Then  $\mathbf{x}^*$  is a local minimum point of  $f(\mathbf{x})$  subject to  $\mathbf{h}(\mathbf{x}) = 0$  and  $\mathbf{g}(\mathbf{x}) \leq 0$ .  $\square$